

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
PART TEST – II
PAPER –1
TEST DATE: 12-12-2021

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 – 04, 20 – 23, 39 – 42): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
Section – A (05 –10, 24 – 29, 43 – 48): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

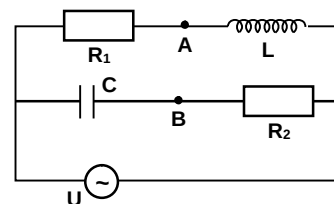
Physics

PART – I

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. The circuit consists of a capacitor, inductance, and two resistors, see figure. The voltage across both the resistors is 10V, and the voltage between the points A and B is also 10V. Find the applied voltage U (in volts).

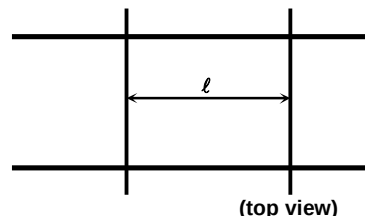


- (A) 10
(B) 20
(C) 30
(D) 40

Ans. B

Sol. Notice that the four vectors form a quadrilateral, opposing angles A and B of which are right angles, hence this is an inscribed quadrilateral, and the other diagonal (other than AB) is the diameter. Pay attention to the fact that the quadrilateral is not convex, because the direction of the voltage vector on L is obtained from that of R_1 by a 90-counter-clockwise rotation (multiplication by $iL\omega/R_1$), and the direction of the voltage vector on R_2 is obtained from that of C by the same rotation (multiplication by $iC\omega R_2$). The problem simplifies further owing to the fact that two sides and one diagonal of the inscribed quadrilateral are all equal to each other.

2. Two identical conducting bars rest on two horizontal parallel conducting rails. The bars are perpendicular to the rails and parallel to each other as shown. The distance between the bars is ℓ . At a certain moment, a uniform vertical upward magnetic field is turned on. The field quickly reaches its maximum magnitude and then remains constant. Neglecting friction, find the new distance between the bars. Assume that the resistance of each bar is much greater than the resistance of the rails.



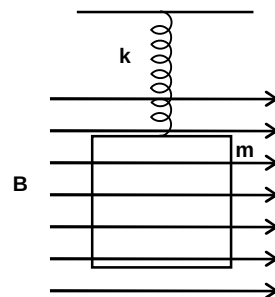
- (A) 2ℓ
(B) 0.5ℓ
(C) ℓ
(D) 0.7ℓ

Ans. B

Sol. First, it should be noted immediately that a trivial dimensional analysis reveals that the answer can only be zero or a numerical multiple of ℓ . It's not hard to guess that the answer might well be $\ell/2$. The somewhat harder part is proving it. this result might well have been anticipated from a combination of

- (1) the dimensional analysis argument,
- (2) the fact that the rods clearly get closer together, and
- (3) an analogy with other problems in which reaching a new equilibrium involves energy dissipation. For instance, when a fully charged capacitor is connected to an identical uncharged capacitor, half the energy is lost and the final potential difference is half the original value. This result depends on there being resistance in the system. Without it we get LC oscillations with each capacitor's potential difference oscillating about half the original value. Similarly, in this problem the resistance is vital in order to allow the magnetic flux through the loop to change from its original value, zero. If the rods had moved to conserve the flux, they would have ended up at a separation of zero, with the resistance they moved just half that far, and ended up with half the flux they would have had if they hadn't moved at all. The answer doesn't depend on R as long as there is some R.

3. A thin copper plate of mass m has a shape of a square with a side b and thickness d . The plate is suspended on a vertical spring with a force constant k in a uniform horizontal magnetic field B parallel to the plane of the plate. Find the period of the small-amplitude vertical oscillations of the plate.



- (A) $2\pi\sqrt{\frac{m + \epsilon_0 b^2 B^2 d}{k}}$
- (B) $2\pi\sqrt{\frac{m}{k}}$
- (C) $2\pi\sqrt{\frac{m + 2\epsilon_0 b^2 B^2 d}{k}}$
- (D) $2\pi\sqrt{\frac{m + 4\pi\epsilon_0 b^2 B^2 d}{k}}$

Ans. A

Sol. Use the concept of motional emf

4. A planet of radius r_0 is at a distance r from the sun ($r \gg r_0$). The sun has radius R . Temperature of the planet is T_0 , and that of the surface of the sun is T_s . Calculate the temperature of another planet whose radius is $2r_0$ and which is at a distance $2r$ from the sun. Assume that the sun and the planets are black bodies.

- (A) $\sqrt{2}T_0$

(B) $\frac{T_0}{\sqrt{2}}$

(C) $2T_0$

(D) $3T_0$

Ans. B

 Sol. Power radiated by the sun = $4\pi\sigma R^2 T_S^4$

Intensity of the sun light at a distance r from the sun is $I = \frac{4\pi\sigma R^2 T_S^4}{4\pi r^2} = \frac{\sigma R^2 T_S^4}{r^2}$

Power received by the planet = $\pi r_0^2 \frac{\sigma R^2 T_S^4}{r^2}$

 If T_0 = temperature of the planet then power radiated by the planet at T_0 must be equal to power received.

$$\pi r_0^2 \frac{\sigma R^2 T_S^4}{r^2} = 4\pi\sigma r_0^2 T_0^4$$

$$\Rightarrow T_0^4 = \frac{R^2}{4r^2} T_S^4$$

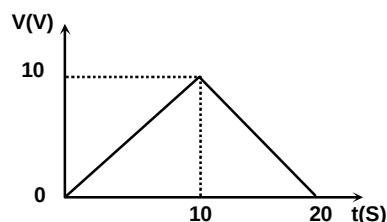
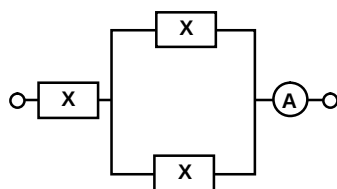
Temperature of the planet does not depend on its radius

 If distance r is doubled, the temperature will become $\frac{T_0}{\sqrt{2}}$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. Element X in the circuit below has a resistance R_X which depends on the voltage V_X on it: for $V_X \leq 1V$, $R_X = 1\Omega$, and for $V_X > 1V$, $R_X = 2\Omega$. Three such elements are connected with an ideal ammeter as shown below; the voltage on the leads of the circuit varies in time as shown in the graph.



Which of the following combinations of (current, time) is/are possible?

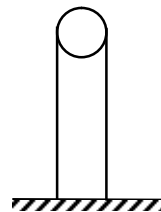
- (A) (1A, 1.5s)
 (B) (2A, 5s)
 (C) (5/3A, 5s)
 (D) (0.6A, 1.5s)

Ans. A, B, C, D

Sol. For the first half of the process, the transitions of the states of the resistors will take place at the overall voltage values $V_1 = 1.5V$ and $V_2 = 5V$; for the second half, the respective transition voltages are $V_2 = 3V$ and $V_1 = 1.25V$. The current time graph will be straight lines connecting the following points:

$(0A, 0s), (1A, 1.5s), (0.6A, 1.5s), (2A, 5), (5/3A, 5s), (10/3A, 10s), (1A, 17s), (1.2A, 17s), (0.5A, 18.75s), (5/6A, 18.75s), (0A, 20s)$

6. There is a long vertical tube of radius r containing air at atmospheric pressure. A steel ball is held at the mouth of the tube and dropped. The ball has radius r and it just fits inside the tube. The tube wall is perfectly smooth and no air can leak from the tube as the ball falls inside it. The ball falls through half the length of the tube before coming to rest. Assume that wall of the tube is perfectly conducting and temperature of the air inside the tube remains constant. Density of steel is d and atmospheric pressure is P_0 and take $L \gg r$. Take air to be an ideal gas.



- (A) The radius (r) of the tube is $\frac{3P_0}{2dg} \ln 2$
- (B) The radius (r) of the tube is $\frac{3P_0}{dg} \ln 2$
- (C) Depth from the top of the tube the ball will be in equilibrium is $L/2$
- (D) Depth from the top of the tube the ball will be in equilibrium is $L \left(1 - \frac{1}{2 \ln 2}\right)$

Ans. A, D

Sol. For isothermal compression of air
 $PV = P_0 V_0$

$$PA [L - x] = P_0 AL = P = P_0 \left(\frac{L}{L - x} \right)$$

$$\text{Work done by the gas on the ball} = -P_0 V_0 \ln \left(\frac{L}{L - x} \right)$$

$$\text{Work done by the gravity on the ball} = mgx$$

$$\text{If the ball comes to rest : } mgx - P_0 V_0 \ln \left(\frac{L}{L - x} \right) = 0$$

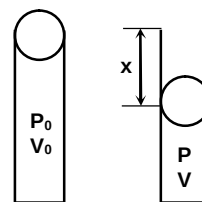
$$\text{Given } x = L/2$$

$$mg \frac{L}{2} = P_0 AL \ln \left(\frac{L}{L - \frac{L}{2}} \right)$$

$$\therefore \frac{4}{3} \pi r^3 dg \frac{1}{2} = P_0 \pi r^2 \ln 2$$

$$\therefore r = \frac{3 P_0 \ln 2}{2 dg}$$

- (b) In equilibrium,
 $P \pi r^2 = mg$



$$P_0 \left(\frac{L}{L-x} \right) \pi r^2 = \frac{4}{3} \pi r^3 dg$$

$$x = L \left(1 - \frac{1}{2 \ln 2} \right)$$

7. An ideal gas is inside a cylinder with a piston that can move freely. The walls of the cylinder and piston are non-conducting. The piston is being moved out of the cylinder at a constant speed u .
- (A) Consider a gas molecule of mass m moving with speed v ($\gg u$). It hits the piston elastically at an angle of incidence θ . The loss in kinetic energy of the molecule is $2mu v \cos \theta$.
- (B) If the area of piston is A and pressure of the gas is P , The rate of decreases of molecular Kinetic energy of the gas sample is PAu .
- (C) If $u \gg$ molecular velocities, the rate at which the gas loses its molecular kinetic energy is $2mu \cos \theta$.
- (D) If $u \gg$ molecular velocities, the rate at which the gas loses its molecular kinetic energy is zero.

Ans. A,B,D

- Sol. (A) The velocity component of the molecule parallel to the wall do not change. Velocity component perpendicular to the wall is $v \cos \theta$ and after the collision it becomes $v \cos \theta - 2u$. [since, speed of approach = speed of separation in an elastic collision]
Loss in kinetic energy of the molecule is

$$\frac{1}{2} m (v \cos \theta)^2 - \frac{1}{2} m (v \cos \theta - 2u)^2 = 2mu v \cos \theta$$
- (B) The product $Fu = PAu$ is the rate at which work is done on the piston or the power developed by the expanding gas. This is the rate at which the molecules will lose kinetic energy as they are not receiving energy from any other source.
- (D) Molecules will not collide with the piston in this case. There is no loss in kinetic energy and no change in temperature.

8. A particle is projected at a speed of $u = 40$ m/s in vertically upward direction in a place where exists a horizontal uniform electric field E_0 . The charge of the particle is $(4mg)/(3E_0)$, where m is the mass of the particle.
- (A) The time after projection, when speed of the particle will be least is $\frac{36}{25}$ seconds.
- (B) The time after projection, when displacement of the particle becomes perpendicular to its acceleration is $\frac{72}{25}$ seconds.
- (C) Assuming that the particle has been projected from a great height and the electric field is present in large region, the angle that the velocity of the particle makes with horizontal after a long time is $\tan^{-1} \left(\frac{3}{4} \right)$

- (D) Assuming that the particle has been projected from a great height and the electric field is present in large region, the angle that the velocity of the particle makes with horizontal after a long time is $\tan^{-1}\left(\frac{4}{3}\right)$

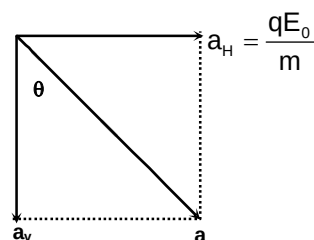
Ans. A, B, C

Sol. Acceleration of the particle
Vertical acceleration $a_v = g$
Horizontal acceleration

$$a_H = \frac{qE_0}{m} = \frac{4gE_0}{3E_0} = \frac{4g}{3}$$

$$a = \sqrt{g^2 + \left(\frac{4g}{3}\right)^2} = \frac{5}{3}g$$

$$\tan \theta = \frac{4}{3}$$



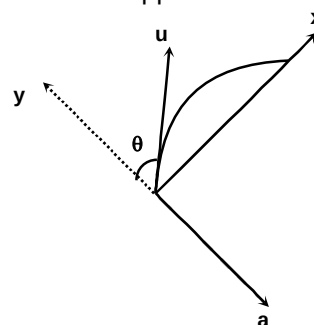
Consider, x as direction perpendicular to acceleration and y as direction opposite to a. Now, the situation is like projectile motion.

$$(a) \ t = \frac{u_y}{a} = \frac{40 \cos \theta}{\frac{5g}{3}} = \frac{40 \times 3 \times 3}{5 \times 10 \times 5} = \frac{36}{25} \text{ s}$$

$$(b) \ 2t = \frac{72}{25} \text{ s}$$

(c) velocity will get almost parallel to acceleration.

$$\therefore \alpha = \tan^{-1}\left(\frac{3}{4}\right)$$



9. A liquid having coefficient of volume expansion γ_0 is filled in a cylindrical glass vessel. Glass has a coefficient of linear expansion of α_g . The liquid along with the container is heated to raise their temperature by ΔT . Mass of the container is negligible.

- (A) For the centre of mass of the system not to move due to heating $\gamma_0 = 2\alpha_g$.
(B) For the centre of mass of the system not to move due to heating $\gamma_0 = 3\alpha_g$.
(C) If the fraction of the volume of the container occupied by the liquid does not change due to heating $\gamma_0 = 2\alpha_g$.
(D) If the fraction of the volume of the container occupied by the liquid does not change due to heating $\gamma_0 = 3\alpha_g$.

Ans. A, D

Sol. The COM will not move if the height of liquid column in the container does not change. Let original volume of liquid, area of cross section of the container and height of liquid be V_0 , A_0 and H_0 respectively.

$$H_0 = \frac{V_0}{A_0}$$

$$V = V_0(1 + \gamma_0 \Delta \theta), \quad A = A_0(1 + 2\alpha_g \Delta \theta)$$

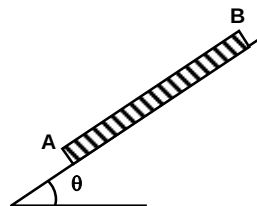
If temperature increases:

$$H = \frac{V_0(1 + \gamma_0 \Delta\theta)}{A_0(1 + 2\alpha_s \Delta\theta)}$$

$$H = H_0$$

$$\therefore \gamma_0 = 2\alpha_g$$

10. A uniform metal rod (AB) of mass m and length L is lying on a rough incline. The inclination of the incline and coefficient of friction between the rod and the incline is $\theta = 37^\circ$ and $\mu = 1.0$ respectively. $\tan 37^\circ = \left(\frac{3}{4}\right)$.



- (A) If temperature increases the rod expands. However, there is a point P on the rod which does not move. The distance of this point from the lower end of the rod is $\frac{L}{2} \left[1 + \frac{\tan \theta}{\mu} \right]$.
- (B) If the temperature falls the rod contracts. Once again there is a point Q which does not move. Find distance of Q from the lower end the rod is $\frac{L}{2} \left[1 - \frac{\tan \theta}{\mu} \right]$.
- (C) Repeated expansion and contraction cause the rod to slide down.
- (D) Repeated expansion and contraction will not cause the rod to slide down.

Ans. A, B, C

- Sol. (a) Let the required distance be ℓ_1 . The lower part AP of the rod mass $\frac{m}{L} \ell_1$ and the upper part

BP has mass $\frac{m}{L}(L - \ell_1)$. When heated, the part AP will be moving down and friction on it (f_1) will be up the incline. The part BP will move up and friction on it will be down (say f_2).

For equilibrium of the entire rod

$$f_1 - f_2 = mg \sin \theta$$

$$\mu \frac{m}{L} \ell_1 g \cos \theta - \mu \frac{m}{L} (L - \ell_1) g \cos \theta = mg \sin \theta$$

$$\frac{\mu}{L} (\ell_1 - L + \ell_1) = \tan \theta \Rightarrow 2\ell_1 = L + \frac{L}{\mu} \tan \theta$$

$$\ell_1 = \frac{L}{2} \left[1 + \frac{\tan \theta}{\mu} \right] = 0.875 L$$

- (b) In this case the friction on upper part will be up and that on the lower part will be down. Proceeding in similar way as above we can get

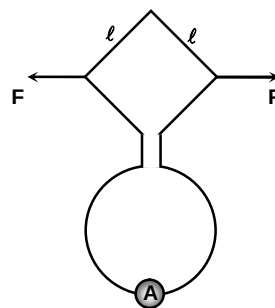
$$\ell_2 = \frac{L}{2} \left[1 - \frac{\tan \theta}{\mu} \right] = 0.125 L$$

- (c) During expansion P is fixed but Q moves down. Then during contraction Q remains fixed. Therefore, in one cycle the point Q has moved down.

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

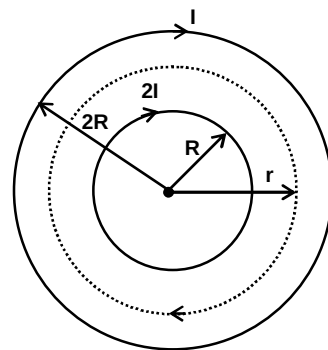
11. The plane of a square loop of wire with edge length $l = 0.2$ m is perpendicular to the Earth's magnetic field at a point where $B = 25 \mu\text{T}$, as shown in figure. The total resistance of the loop and the wires connecting it to a sensitive ammeter is 0.5Ω . If the loop is suddenly collapsed by horizontal forces as shown, what is the total charge, in μC , that passes through the ammeter?



Ans. 2

Sol. Use Faraday's law.

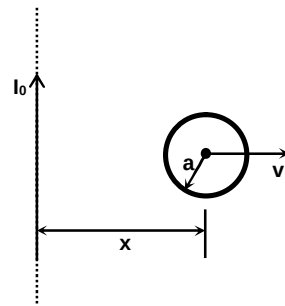
12. A long solenoid of radius $2R$ contains another coaxial solenoid of half the radius (The diagram shows cross section of these solenoids). The coils have the same number of turns per unit length and initially both carry no current. At a same instant, the currents in both solenoids start increasing linearly with time. At any moment the current flowing in the inner coil is twice as large as that in the outer one and their charged particle, initially at rest between the solenoids, starts moving along a circular trajectory (see figure) of radius $r = \sqrt{k}R$. Then find k .



Ans. 2

Sol. The change in current in the solenoid will cause a time varying magnetic field.

13. A small conducting loop for radius a and resistance per unit length λ , is pulled with velocity v perpendicular to a long straight conductor carrying a current I_0 . If a constant power P is dissipated in the loop, the variation of velocity of the loop as a function of x is $\frac{x^2}{\mu_0 I_0 a^2} \sqrt{k a \pi \lambda P}$, where k is an integer. {Given that $x \gg a$ }.



Ans. 8

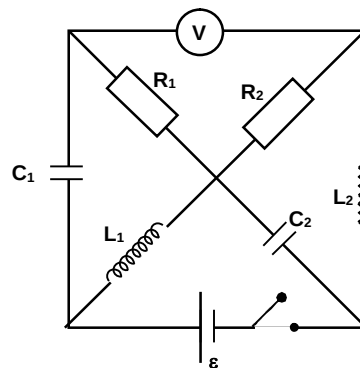
Sol. Use Faraday's law. The magnetic field on the circular loop can be assumed to be approximately constant as $x \gg a$.

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15
Question Stem

For the circuit shown in Figure, $R_1 = 3\Omega$, $R_2 = 1\Omega$, $C_1 = C_2 = 2F$, and $L_1 = L_2 = L = 2H$. The electromotive force of the battery is $\varepsilon = 1\text{V}$. Initially the switch is closed and the system is operating in a stationary regime (steady state).



14. The reading of the voltmeter in the stationary regime is

Ans. 1.00

Sol. The reading will be equal to ε .

15. The total amount of heat (in Joules) which will be dissipated on each of the resistors after opening the switch, and until a new equilibrium state is achieved is

Ans. 2.00

Sol. The total amount of heat which will be dissipated on each of the resistors after opening the switch, and until a new equilibrium state is achieved will be $\frac{C\varepsilon^2}{2} + \frac{L\varepsilon^2}{2R^2} = 2J$

Question Stem for Question Nos. 16 and 17
Question Stem

A spherical black body of radius r at absolute temperature T is surrounded by a thin spherical and concentric shell of radius R , black on both sides. The factor by which this radiation shield reduces the rate of cooling of the body (consider space between spheres evacuated, with no thermal conduction losses) is

given by the following expression: $\frac{aR^2}{R^2 + br^2}$

16. The value of a is

Ans. 1.00

17. The value of b is

Ans. 1.00

Sol. (Q.16-17)

Let the surrounding temperature be T_0 . The rate of energy loss of the black body before being surrounded by the spherical shell is : $Q = 4\pi r^2 \sigma (T^4 - T_0^4)$

The energy loss per unit time by the black body after being surrounded by the shell is $Q' = 4\pi r^2 \sigma (T^4 - T_1^4)$.where T_1 is temperature of the shell.

The energy loss per unit time by the shell is $Q'' = 4\pi r^2 \sigma (T^4 - T_0^4)$

Since $Q'' = Q'$, we obtain $\frac{Q'}{Q} = \frac{R^2}{R^2 + r^2}$

Question Stem for Question Nos. 18 and 19

Question Stem

Two small positively charged spheres are suspended from a common point at the ceiling by the insulating light strings of equal length. The first sphere has mass m_1 and charge q_1 while the second one has mass m_2 and charge q_2 . If the first string makes an angle θ_1 with the vertical, and the second string makes an angle θ_2 with the vertical.

18. Find the value of θ_1 (in degrees) if $m_1 = m_2$, $q_1 = 2q_2$ and $\theta_2 = 30^\circ$

Ans. 30.00

19. Find the value of θ_1 (in degrees) if $m_2 = \sqrt{2}m_1$, $q_1 = 13q_2$ and $\theta_2 = 45^\circ$

Ans. 30.00

Sol. (Q.18-19)

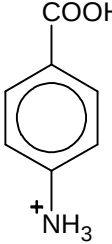
Use the condition for equilibrium to prove that $\theta_2 = \sin^{-1} \left(\frac{m_1}{m_2} \sin \theta_1 \right)$

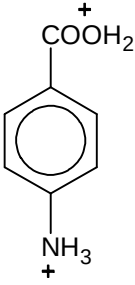
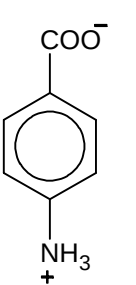
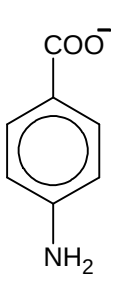
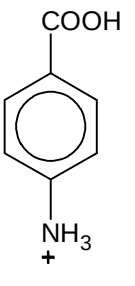
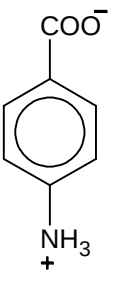
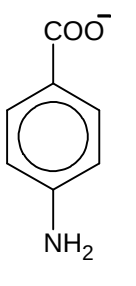
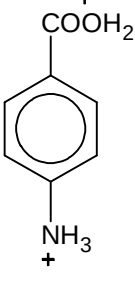
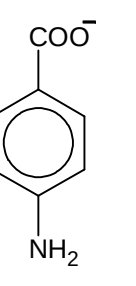
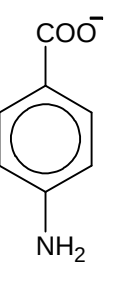
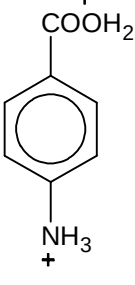
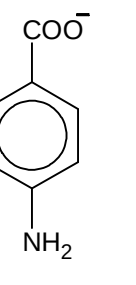
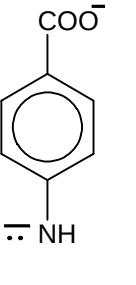
Chemistry

PART – II

Section – A (Maximum Marks: 12)

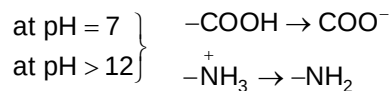
This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

20. Major species obtained when  is dissolved in H₂O at pH = 2, pH = 7 and pH = 12, respectively are? $pK_{a_1} = 2.35, pK_{a_2} = 4.98$.

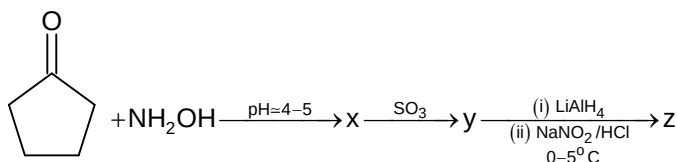
- (A)  ,  , 
- (B)  ,  , 
- (C)  ,  , 
- (D)  ,  , 

Ans. C

Sol. At $\text{pH} \leq 2$ both COOH and NH_2
Remains protonated



21.

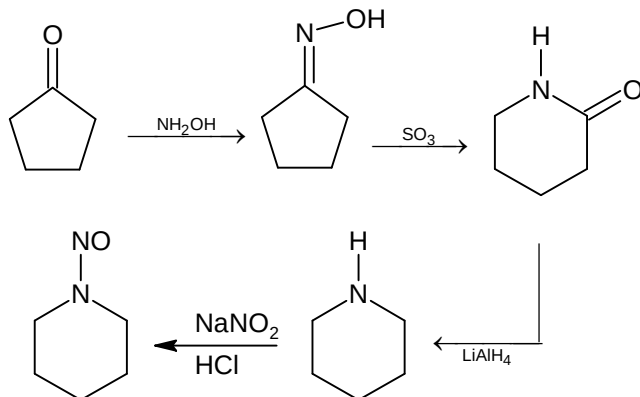


Which option is correct?

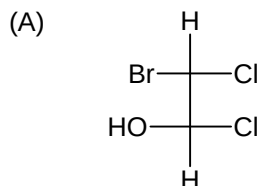
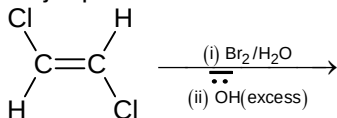
- (A) x is α -amino ketone
- (B) y is cyclic amide
- (C) z is yellow oily liquid
- (D) both (B) and (C) are correct

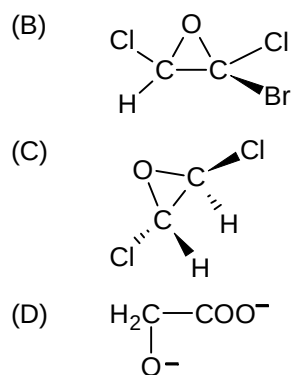
Ans. D

Sol.



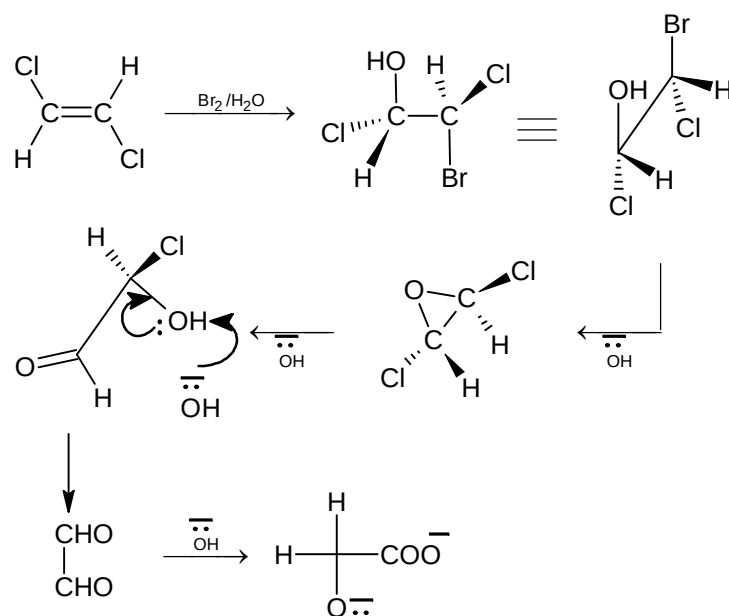
22. Major product of reaction is



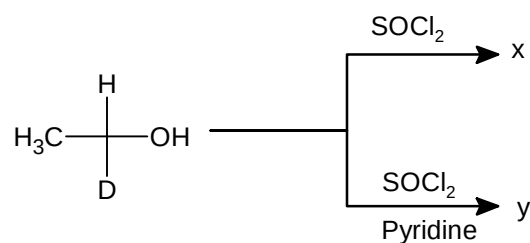


Ans. D

Sol.



23.



Which statement is correct?

- (A) x and y are superimposable mirror images
- (B) x and y are non-superimposable mirror images
- (C) x and y are neither mirror images nor super imposable
- (D) x and y are structural isomers

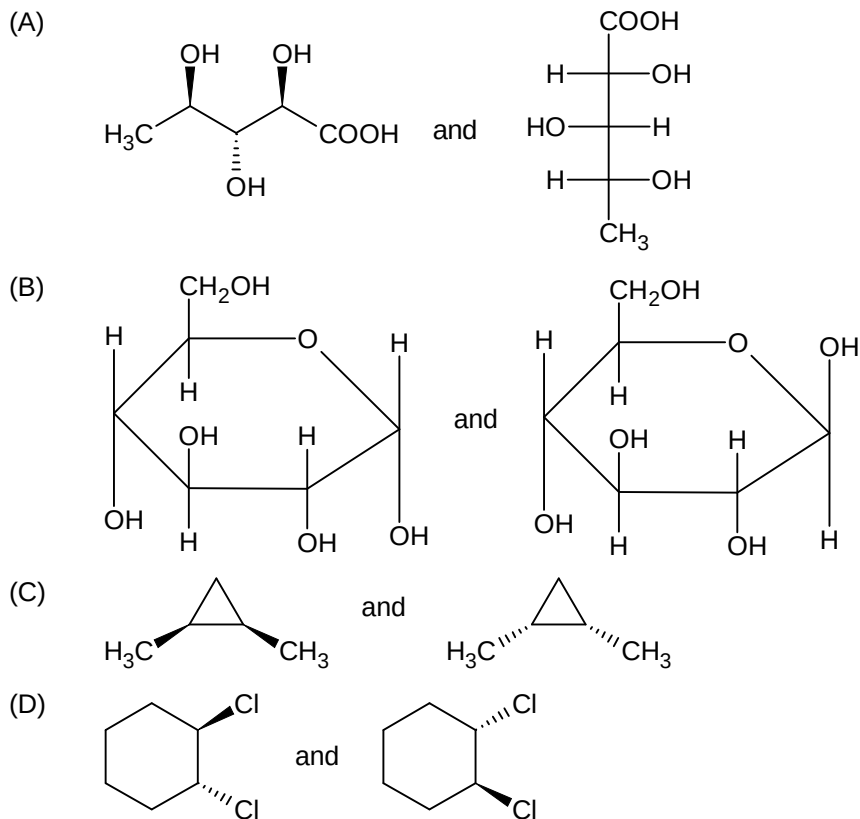
Ans. B

Sol. x is formed via S_N1 and y is formed via S_N2 .

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

24. Which of the following pair of compounds does not represent enantiomeric pair?



Ans. A, B, C

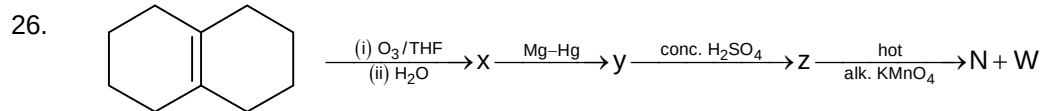
Sol. A – diastereomeric pair.
B – diastereomeric pair.
C – identical pair.

25. Which of the following does not give Lassaigne's test for nitrogen?

- (A) Hydrazine
(B) HN_3
(C) $\text{PhN}_2^+ \text{Cl}^-$
(D) CH_3CONH_2

Ans. A, B, C

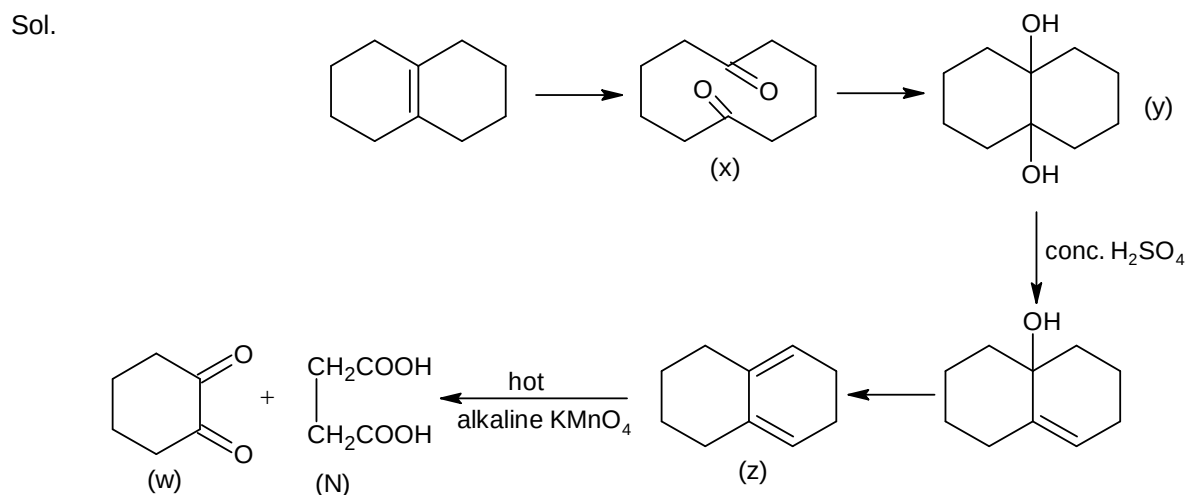
Sol. $\text{PhN}_2^+ \text{Cl}^-$ gives off N_2 while N_2H_4 and HN_3 don't have 'C' atoms to form NaCN .



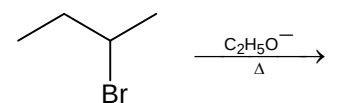
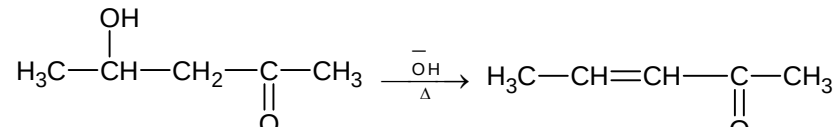
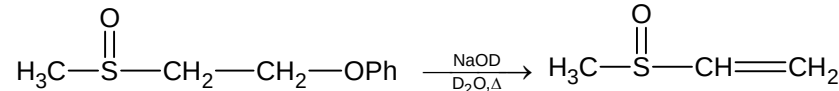
Which of the following option(s) is/are correct?

- (A) x is a diketone
- (B) y is a gemdiol
- (C) z is a conjugated diene
- (D) W and N are β -diketone and β -dicarboxylic acid

Ans. A, C, D



27. How many of the following elimination takes place via E_{ICB} mechanism

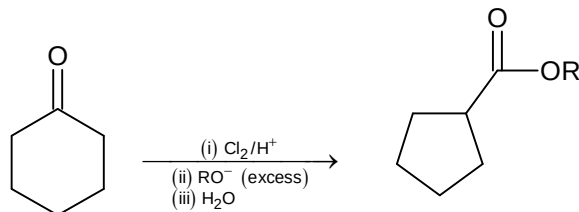
- (A) 
- (B) 
- (C) 
- (D) 

Ans. B, C, D

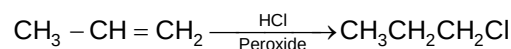
Sol. Carbanion is stable in B, C, D.

28. Which of the following reactions/statements is/are correct?

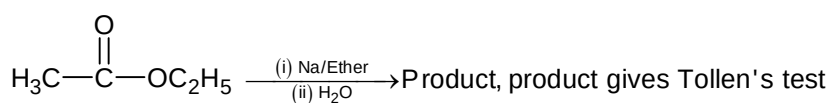
(A)



(B)



(C)



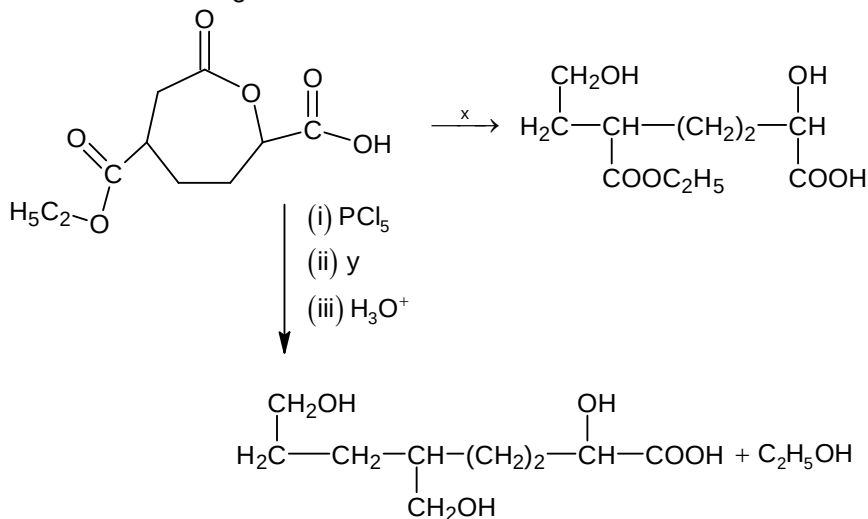
(D)



Ans. A, C, D

Sol. Only HBr shows peroxide effect. α - hydroxy acid forms Lactide on heating.

29. Select the best reagent for the conversion



(A) x is LiAlH_4

(B) x is NaBH_4

(C) y is NaBH_4

(D) y is B_2H_6 / THF

Ans. B, D

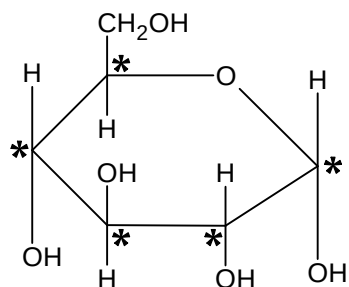
Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. Number of chiral carbon atoms in α – D (+) glucose?

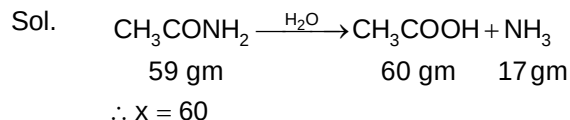
Ans. 5

Sol.



31. 59 gm of an amide is obtained from a carboxylic acid. Which on hydrolysis gives 17 gm of NH_3 , the molecular mass of acid is x. x is.

Ans. 60



32. Platinum salt of an organic base contains 25% platinum. Equivalent weight of the base is E. The value of E is [Atomic mass of platinum is 195 and atomic mass of Cl is 35.5].

Ans. 185

Sol.

$$E = \frac{1}{2} \left[\left(\frac{100}{25} \right) \times 195 - 410 \right] = 185$$

$E = 185$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34
Question Stem

0.35 gm of an organic compound containing carbon, hydrogen and oxygen was analysed by the combustion method. The increase in mass of calcium chloride tube and potash bulbs at the end of the experiment was found to be 0.21 gm and 0.26 gm respectively. On analysis, the above organic compound is found to contain x % carbon and y % oxygen by mass, respectively.

33. Value of x is _____

Ans. 20.26
(Range : 20.25 – 20.27%)

34. Value of y is _____

Ans. 73.07
(Range: 73.00 – 73.10%)

Sol. (for the Q. No. 33 and 34)

$$\% \text{ of C} = \frac{12}{44} \times \frac{0.26}{0.35} \times 100 = 20.2597 \approx 20.26$$

$$\% \text{ of H} = \frac{1}{18} \times \frac{0.21}{0.35} \times 100 = 6.6666 \approx 6.67$$

$$\% \text{ oxygen} = 100 - (20.26 + 6.67) = 73.07$$

Question Stem for Question Nos. 35 and 36

Question Stem

The rate law for the substitution reaction of 2-bromobutane and OH^- in 75% ethanol – 25% water at 30°C is rate = $3.0 \times 10^{-5} [2\text{-bromobutane}] [\text{OH}^-] + 1.5 \times 10^{-6} [2\text{-bromobutane}]$

35. What percent of the reaction takes place by the $\text{S}_{\text{N}}2$ mechanism when $[\text{OH}^-] = 1.0 \text{ M}$?

Ans. 95.24
(Range : 95.23 – 95.25)

Sol.
$$\% \text{ of } \text{S}_{\text{N}}2 = \frac{3.0 \times 10^{-5} [2\text{-bromobutane}]}{3.0 \times 10^{-5} [2\text{-bromobutane}] + 1.5 \times 10^{-6} [2\text{-bromobutane}]} \times 100$$

$$= 95.238 \approx 95.24$$

36. What percentage of reaction takes place by the $\text{S}_{\text{N}}2$ mechanism when $[\text{OH}^-] = 0.01 \text{ M}$?

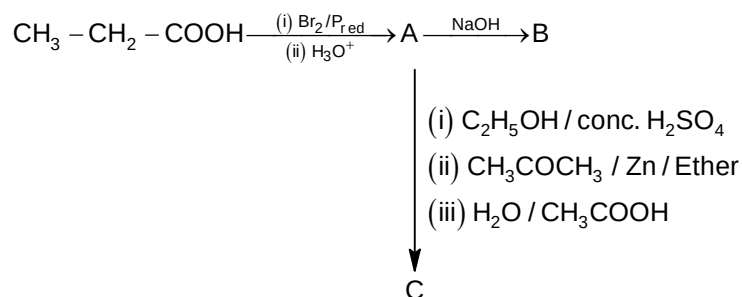
Ans. 16.67
(Range: 16.65 – 16.68)

Sol.
$$\% \text{ of } \text{S}_{\text{N}}2 = \frac{(3.0 \times 10^{-5})(0.01)[2\text{-bromobutane}]}{3.0 \times 10^{-5} [2\text{-bromobutane}] + 1.5 \times 10^{-6} [2\text{-bromobutane}]} \times 100$$

$$= 16.67$$

Question Stem for Question Nos. 37 and 38

Question Stem



[Given atomic mass of C = 12, H = 1, O = 16, Na = 23]

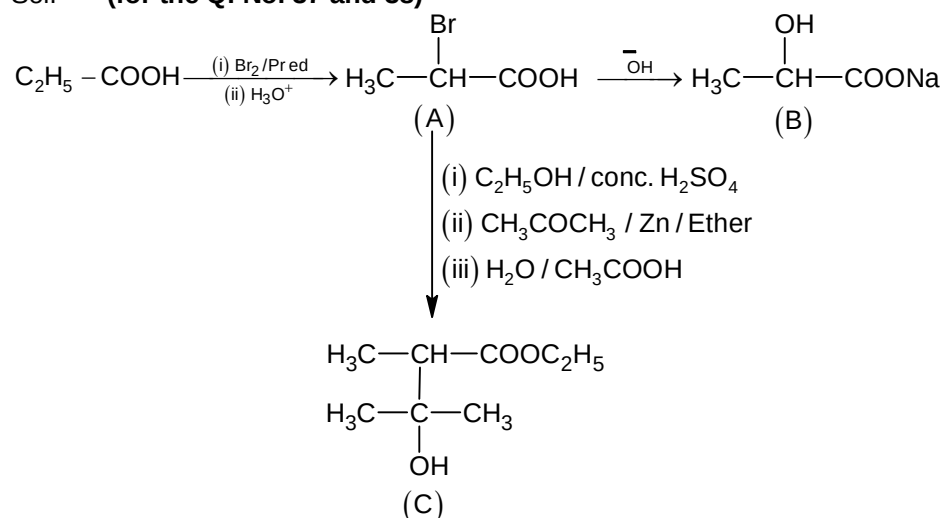
37. If 7.4 gm of $\text{CH}_3\text{CH}_2\text{COOH}$ is taken, then the mass of B (in gm) produced will be?

Ans. 11.20

38. Mass of 'C' (in gm) produced by 3.6 gm of $\text{C}_2\text{H}_5\text{COOH}$ is?

Ans. 7.78
(Range: 7.77 – 7.79)

Sol. (for the Q. No. 37 and 38)



Molar mass of $\text{C}_2\text{H}_5\text{COOH} = 74 \text{ gm}$

Molar mass of $\text{CH}_3\text{CH}(\text{OH})\text{COONa} = 112$

$\therefore 74 \text{ gm}$ gives $\longrightarrow 112 \text{ gm}$

$\therefore 7.4 \text{ gm}$ gives $\longrightarrow \frac{112}{74} \times 7.4 = 11.2 \text{ gm}$

$\therefore 3.6 \text{ gm C}_2\text{H}_5\text{COOH}$ gives $= \frac{160}{74} \times 3.6$
 $= 7.7837$
 ≈ 7.78

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

39. Let ABC be an isosceles triangle with $AB = AC = 6$ units and let D be mid-point of AC. A circle through D touches AB at B. If $OC = 2$, then radius of the circle is equal to

- (A) 4
(B) $3\sqrt{2}$
(C) $2\sqrt{5}$
(D) $\sqrt{22}$

Ans. D

- Sol. AD is extended to meet the circle at E and OC is extended to meet the circle at H and F

Clearly, $AD \cdot AE = AB^2$

$$\Rightarrow 3 \cdot (6 + CE) = 6^2 \Rightarrow CE = 6$$

Again $CD \cdot CE = CF \cdot CH$

$$\Rightarrow 3 \cdot 6 = (r + 2)(r - 2) \Rightarrow r = \sqrt{22}$$

40. If $\tan^{-1}|a| = \tan^{-1}(a^2 + k) - \frac{\pi}{4}$ holds for integral values of $a \in [-2, 2]$. Then number of integral value(s) of k is equal to

- (A) 1
(B) 2
(C) 3
(D) 0

Ans. A

- Sol. Given equation $\Rightarrow \tan^{-1}(a^2 + k) = \tan^{-1}|a| + \frac{\pi}{4}$

Case (1): If $|a| = 0 \Rightarrow k = 1$

Case (2): If $|a| = 1 \Rightarrow k \in \phi$

Case (3): If $|a| = 2 \Rightarrow \tan^{-1}|a| + \frac{\pi}{4} = \tan^{-1}(a^2 + k)$

$$\Rightarrow \pi + \tan^{-1} \frac{|a| + 1}{1 - |a|} = \tan^{-1}(k + a^2)$$

$$\Rightarrow \pi - \tan^{-1} 3 = \tan^{-1}(k + 4); \text{ not possible as } \text{LHS} \in \left(\frac{\pi}{2}, \pi\right) \text{ and } \text{RHS} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

41. Let $S \equiv 0$ and $S' \equiv 0$ be two circles belonging to the family of circles $x^2 + y^2 + \lambda x - 4y + 3 = 0 : (\lambda \in \mathbb{R})$ such that $S \equiv 0$ bisects the circumference of circle $S' \equiv 0$ and has radius equal to twice of radius of $S' \equiv 0$. Then distance between their centres is

- (A) $\frac{1}{2}$
 (B) 1
 (C) 2
 (D) $\frac{1}{\sqrt{2}}$

Ans. B

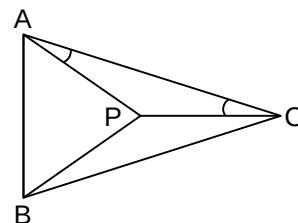
Sol. Given family of circles is $x^2 + y^2 - 4y + 3 + \lambda x = 0$
 $\Rightarrow$ Common points are (0, 1) and (0, 3)
 Clearly $S' \equiv 0$ will be the circle with common points as diameter
 $\Rightarrow S' \equiv (x - 0)(x - 0) + (y - 1)(y - 3) = 0$
 $\Rightarrow x^2 + y^2 - 4y + 3 = 0$
 $\Rightarrow$ Centre $S' \equiv 0$ is (0, 2) and radius = 1
 $\Rightarrow$ Radius of $S \equiv 0$ is $2 = \sqrt{\frac{\lambda^2}{4} + 4 - 3} \Rightarrow \lambda = \pm 2$
 $\Rightarrow$ Distance between their centres = 1 unit

42. Let P be a point inside a triangle ABC such that $PA = PC = AB$ and angles ABC, PAC and PCB are of magnitude 12α , 3α and 2α respectively. Then maximum value of expression $f(\theta) = \sin 5\alpha \sin \theta + \cos 10\alpha \cos \theta$, ($\theta \in \mathbb{R}$) is equal to

- (A) $\sqrt{2}$
 (B) 1
 (C) 2
 (D) $\frac{1}{\sqrt{2}}$

Ans. D

Sol. We have $\angle PAB = 180^\circ - (5\alpha + 12\alpha + 3\alpha) = 180^\circ - 20\alpha$
 As $\angle PBA = \angle BPA = \beta$ (say) $\Rightarrow 180^\circ - 2\beta = 180^\circ - 20\alpha \Rightarrow \beta = 10\alpha$
 $\Rightarrow \angle PBC = 2\alpha \Rightarrow PB = PC \Rightarrow P$ is circumcentre
 $\Rightarrow \angle APC = 2\angle ABC = 24\alpha \Rightarrow 24\alpha + 3\alpha + 3\alpha = 180^\circ \Rightarrow \alpha = 6^\circ$
 We get $f(\theta) = \frac{1}{2}(\sin \theta + \cos \theta) \Rightarrow$ Maximum value = $\frac{1}{\sqrt{2}}$



Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

43. The chord AB of the circle $x^2 + y^2 = 8$ is given by $mx - y = 2m - 2$; $m \in \mathbb{R}$. If P is a variable point on the circle such that $\angle APB = \frac{\pi}{4}$; then

- (A) there exist exactly two such chords
- (B) there exist exactly one such chord
- (C) slope of chord is 1
- (D) length of chord is 4 units

Ans. A, D

Sol. Clearly, variable chord passes through the fixed point (2, 2) lying on the circle $\angle APB = \frac{\pi}{4}$

$$\Rightarrow \angle AOB = \frac{\pi}{2}$$

$\Rightarrow$ Chord subtends right angle at the origin

$$\text{Now on homogenization, we get } x^2 + y^2 - 8 \left(\frac{mx - y}{2m - 2} \right)^2 = 0$$

$$\Rightarrow (m - 1)^2 [x^2 + y^2] - 2[m^2x^2 - 2mxy + y^2] = 0$$

$$\Rightarrow \text{Coefficient of } x^2 + \text{coefficient of } y^2 = 0$$

$$\Rightarrow (m - 1)^2 - 2m^2 + (m - 1)^2 - 2 = 0$$

$$\Rightarrow 0m^2 - 4m + 0 = 0 \Rightarrow m = 0 \text{ and undefined (A) and (D) are correct}$$

44. Let $C_1 : x^2 + y^2 - 60x - 20y + k_1 = 0$ and $C_2 : x^2 + y^2 - 20x - 30y + k_2 = 0$. If ABC is a triangle with its vertices on C_1 . Let altitudes AD, BE and CF intersect C_1 again at points H, J and K respectively. If C_2 is incircle of $\triangle DEF$ and $D \equiv (8, 9)$, then which of the following is/are TRUE?

- (A) K_1 is equal to 375
- (B) The distance of orthocentre of $\triangle ABC$ from vertex C is equal to $5\sqrt{2}$ units
- (C) AD is equal to $7\sqrt{10}$ units
- (D) Area of hexagon AKBHCJA is equal to 1050 sq. units

Ans. A, B, C, D

Sol. Clearly incentre of triangle DEF is orthocentre of $\triangle ABC$

$\Rightarrow$ Orthocentre $P \equiv (10, 15)$ and as mirror image of orthocentre about any side of triangle lies on the circumcircle : equation of side BC $\perp$ DP is $x + 3y - 35 = 0$

$$\Rightarrow H \equiv (6, 3) \Rightarrow \text{radius of circumcircle} = 25 \text{ units} \Rightarrow K_1 = 375$$

$$\Rightarrow \text{Equation of circumcircle is } (x - 30)^2 + (y - 10)^2 = 25^2 \quad \dots (1)$$

$$\text{Equation of side BC is } x + 3y - 35 = 0 \quad \dots (2)$$

$$\text{and equation of altitude AD is } 3x - y - 18 = 0 \quad \dots (3)$$

Solving equation (1), (2) and (3), we get A, B and C as (15, 30), (50, -5) and (5, 10)

$$\Rightarrow CP = 5\sqrt{2} \text{ and } AD = 7\sqrt{10}$$

$$\text{Area of hexagon} = 2\Delta ABC = AD \cdot BC = 1050 \text{ sq. units}$$

45. Let a line parallel to the axis of the parabola $y^2 = 4x$ be drawn through point $P(\alpha - 5, \alpha)$ to meet the parabola at Q. Equation of tangent at Q is $x - 2y + 4 = 0$. Let T be a point on the tangent, and M, N be feet of perpendiculars on SQ and directrix respectively from point T, S is focus, then

- (A) $\alpha (\neq 0)$ is equal to -4
 (B) if $QM = 3$, then TN is 2
 (C) if two tangents are drawn from point P to the parabola touching it at A and B and circumcentre of ΔPAB is at (β, γ) , then γ is -1
 (D) if orthocentre of ΔPAB (as defined in option C) is at (a, b) , then $a + b$ equals to 3

Ans. B, D

Sol. Let $Q \equiv \left(\frac{\alpha^2}{4}, \alpha\right)$ lies on $x - 2y + 4 = 0$

$$\Rightarrow \alpha = 4$$

Δ s TMQ and TRQ are congruent

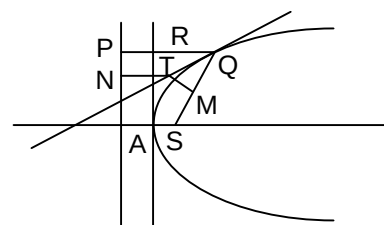
$$\Rightarrow QM = QR$$

$$\Rightarrow SM = TN = SQ - QM = 5 - 3 = 2$$

Now, $P = (-1, 4)$ which lies on directrix

$\Rightarrow$ Two tangents are perpendicular

$\Rightarrow$ P is orthocentre and mid-point of AB as circumcentre which has same y co-ordinate as that of P



46. Let $S = \sin^4 12^\circ + \left[\frac{\sin^2 24^\circ}{2}\right]^2 + \left[\frac{\sin^2 48^\circ}{4}\right]^2 + \left[\frac{\sin^2 96^\circ}{8}\right]^2 + \dots$ upto 9 terms.

If $P = [2^9 \operatorname{cosec} 12^\circ]^2 \times S$, then P is divisible by

- (A) 3
 (B) 4
 (C) 7
 (D) 21

Ans. A, C, D

Sol. We have $\sin^4 \theta = \sin^2 \theta (1 - \cos^2 \theta) = \sin^2 \theta - \frac{1}{4} \sin^2 2\theta$

Putting $\theta = 12^\circ, 24^\circ, 48^\circ, 96^\circ, \dots$ etc., we get

$$\sin^4 12^\circ = \sin^2 12^\circ - \frac{1}{4} \sin^2 24^\circ$$

$$\sin^4 24^\circ = \sin^2 24^\circ - \frac{1}{4} \sin^2 48^\circ$$

.....

.....

Putting the values in S, we get

$$S = \sin^2 12^\circ - \frac{1}{4^9} \sin^2 3072^\circ = \frac{4^9 - 1}{4^9} \sin^2 12^\circ$$

$$\Rightarrow P = 4^9 - 1$$

47. The tangents $x - 2 = 0$ and $x - 2y + 10 = 0$ drawn to a parabola intersect the tangent at the vertex at $(2, 0)$ and $(-2, 4)$ respectively, then

- (A) equation of tangent at the vertex is $x + y - 2 = 0$
 (B) focus is $(0, -1)$
 (C) equation of axis of parabola is $x - y - 1 = 0$
 (D) length of latus rectum is $4\sqrt{2}$ units

Ans. A, D

Sol. As perpendicular from focus on any tangent to a parabola meets it on the tangent at vertex
 $\Rightarrow$ Focus is point of intersection of lines perpendicular to the given tangents, we get the lines as $y = 0$ and $2x + y = 0$
 $\Rightarrow$ Focus $\equiv (0, 0)$. Again axis is perpendicular to tangent at the vertex and passes through focus
 $\Rightarrow$ Equation is $y = x$

48. Let P be a point on the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, in first quadrant with focal distance 7 units w.r.t focus $S'(-4, 0)$. Let $S'P$ and perpendicular from focus S on the tangent at P intersect at Q. Then which of the following is not TRUE?

- (A) PQ is equal to 3
 (B) SQ is equal to $\frac{3\sqrt{3}}{\sqrt{7}}$
 (C) Slope of tangent at P is equal to $\frac{\sqrt{3}}{5}$
 (D) Let P be a variable point then locus of Q is a circle with radius = 10 units

Ans. B, C

Sol. Clearly $S \equiv (4, 0)$; Let $P = (\alpha, \beta)$

$$\Rightarrow a + e\alpha = 7 \Rightarrow 5 + \frac{4}{5}\alpha = 7 \Rightarrow \alpha = \frac{5}{2} \text{ and } \beta = \frac{3\sqrt{3}}{2}$$

$$\text{Hence, equation of tangent at P is } \sqrt{3}x + 5y - 10\sqrt{3} = 0$$

As tangent is bisector of angle between SP and $S'P$

$$\Rightarrow PQ = PS = 3. \text{ As } PS + PS' = 10 = \text{Length of major axis}$$

$$\text{When P is variable then } S'Q = S'P + PQ = S'P + PS = 10$$

$$\Rightarrow \text{Variable point Q is at a fixed distance from } S'$$

$$\Rightarrow \text{Locus is a circle with centre } S' \text{ and radius} = 10 \text{ units}$$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. Let ABC be an isosceles triangle obtuse at A. If two altitudes are of lengths 3 and 2 respectively, then area of ΔABC is k, then $[k]$ is equal to (where $[.]$ denotes greatest integer function)

Ans. 4

Sol. Clearly altitudes are of lengths 2, 3 and 3

$$\text{Also we have } \Delta = \frac{1}{2}a \cdot AD = \frac{1}{2}b \cdot BE = \frac{1}{2}c \cdot CF$$

$$\Rightarrow a = \frac{2\Delta}{2}; b = \frac{2\Delta}{3} \text{ and } c = \frac{2\Delta}{3}$$

$$\Rightarrow S = \frac{a+b+c}{2} = \frac{\Delta}{2} + \frac{\Delta}{3} + \frac{\Delta}{3} = \frac{7}{6}\Delta$$

$$\Rightarrow \Delta = \sqrt{\frac{7\Delta}{6} \cdot \frac{\Delta}{2} \cdot \frac{\Delta}{2} \cdot \frac{\Delta}{6}} \Rightarrow \Delta = \frac{12}{\sqrt{7}} = 4.54 \text{ sq. units}$$

50. Let H : $xy - y - 2x - 2 = 0$. If normal drawn to H at $P(3, 4)$ intersects the curve again at $Q(\alpha + 1, \beta + 1)$, then $\frac{\alpha}{\beta}$ is

Ans. 2

Sol. Given equation of curve $(x - 1)(y - 2) = 4$; hyperbola

$$\text{Let } x - 1 = X \text{ and } y - 2 = Y \Rightarrow XY = 4$$

Clearly normal at $\left(ct, \frac{c}{t}\right)$ intersects the curve again at $\left(ct', \frac{c}{t'}\right)$ where $t' = -\frac{1}{t^3}$; $c = 2$

$$\Rightarrow 2t = 2 \Rightarrow t = 1 \Rightarrow Q = (-2, -2) \Rightarrow Q \text{ in old system } (\alpha + 1, \beta + 1) \equiv (-1, 0)$$

$$\Rightarrow \frac{\alpha}{\beta} = 2$$

51. Number of solutions of equation $|\sin \alpha| = 2d$ ($\alpha \in [-2\pi, 2\pi]$) where d represents the shortest distance between the curves $2x^2 - 2y + 1 = 0$ and $2y^2 - 2x + 1 = 0$ is equal to

Ans. 8

Sol. Shortest distance lies along common normal

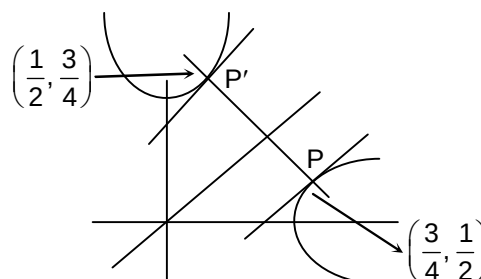
$$\Rightarrow \text{Slope of common normal} = -1$$

$$\text{Now on differentiation of curve } 2y^2 - 2x + 1 = 0$$

$$\Rightarrow 4y \frac{dy}{dx} - 2 = 0 \Rightarrow -\frac{dx}{dy} = -2y = -1$$

$$\Rightarrow y = \frac{1}{2} \Rightarrow d = PP' = \frac{1}{2\sqrt{2}} \Rightarrow |\sin \alpha| = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \text{Number of solution} = 8$$



Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53

Question Stem

Let ABCD be a quadrilateral with $\angle ADB = \frac{\pi}{6}$, $\angle CBD = \frac{\pi}{4}$ and $\angle BDC = \frac{5\pi}{12}$ if $AD = 2AB$; and diagonals intersect at M, then

52. If $\angle CMD = \frac{\pi}{k}$, then k is equal to _____

Ans. 3.00

53. If $AB = \sqrt{6} - \sqrt{2}$, then distance between orthocentre and ex-centre (opposite angle A) of $\triangle ABD$ is equal to _____

Ans. 2.00

Sol. (52.-53.)

We are given $\frac{AB}{AD} = \frac{1}{2}$

$$\Rightarrow \frac{\sin 30^\circ}{\sin \theta} = \frac{1}{2}$$

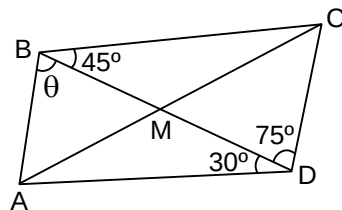
$$\Rightarrow \theta = \frac{\pi}{2}$$

Clearly, BC and DC are external angle bisectors of $\triangle ABD$

$\Rightarrow C$ is ex-centre of $\triangle ABD$

Clearly, B is orthocentre as $\angle ABD = \frac{\pi}{2}$. Now, using sin rule in $\triangle ABC$

$$\frac{AB}{\sin \frac{\pi}{12}} = \frac{BC}{\sin \frac{\pi}{6}} \Rightarrow BC = \frac{(\sqrt{6} - \sqrt{2}) \times \frac{1}{2}}{\frac{\sqrt{3} - 1}{2\sqrt{2}}} = 2$$



Question Stem for Question Nos. 54 and 55

Question Stem

ABCD is a concyclic quadrilateral on the curve $xy - x - y - 3 = 0$. Let E, F and H be feet of perpendiculars from vertices A, B and C on sides BC, CA and AB of $\triangle ABC$. If $A \equiv (0, -3)$, $D \equiv (\beta, \gamma)$ and incentre of $\triangle EFH$ is at (α, α) (in first quadrant). If $\triangle ABC$ and $\triangle BCD$ are acute angle triangles, then

54. The value of $\beta^2 - \gamma$ is equal to _____

Ans. 2.00

55. Distance between orthocentres of $\triangle ABC$ and $\triangle BCD$ is equal to _____

Ans. 2.24

Sol. (54.-55)

As EFH is pedal triangle

$\Rightarrow$ Incentre of $\triangle EFH$ is orthocentre of $\triangle ABC$ which lies on the curve

$$\Rightarrow \alpha^2 - 2\alpha - 3 = 0 \Rightarrow \alpha = 3$$

$$\Rightarrow D \equiv (-1, -1)$$

$$\Rightarrow \beta^2 - \gamma = 1 - (-1) = 2$$

$$\text{Distance between orthocentres} = \text{Distance between A and D} = \sqrt{(0+1)^2 + (-3+1)^2}$$

$$= \sqrt{5} = 2.24$$

Question Stem for Question Nos. 56 and 57

Question Stem

Consider the parabola $E_1 : y^2 = 8x$ and ellipse $E_2 : x^2 + 4y^2 - 4k = 0$. If a common tangent to E_1 and E_2 meets them at points A and B respectively. If eccentric angle of point B is $\frac{2\pi}{3}$; then

56. The value of k is equal to _____

Ans. 36.00

Sol. Equation of tangent having slope m for $y^2 = 8x$ is $y = mx + \frac{2}{m}$

$$\Rightarrow m^2x - my + 2 = 0 \quad \dots (1)$$

Equation of tangent to the ellipse with parametric angle θ is

$$\frac{x \cos \theta}{2\sqrt{k}} + \frac{y \sin \theta}{\sqrt{k}} = 1 \quad \dots (2)$$

For common tangent (1) and (2) are identical

$$\Rightarrow m^2 = \frac{-\cos \theta}{\sqrt{k}} \text{ and } m = \frac{2 \sin \theta}{\sqrt{k}} \Rightarrow \frac{-\cos \theta}{\sqrt{k}} = \frac{4 \sin^2 \theta}{k}$$

$$\Rightarrow \sqrt{k} = -\frac{4 \sin^2 \theta}{\cos \theta} = 6, \text{ for } \theta = \frac{2\pi}{3}$$

$$\Rightarrow k = 36$$

57. If two of the three normals drawn from the point $(\alpha, 0)$ on the ellipse to the parabola $y^2 = 8x$ are perpendicular, then α is equal to _____

Ans. 6.00

Sol. Equation of normal to parabola is $y = mx - 4m - 2m^3$

$$\Rightarrow 2m^3 + (4 - \alpha)m = 0 \Rightarrow m[2m^2 + (4 - \alpha)] = 0$$

$$\text{Product of slopes} = -1 \Rightarrow \frac{4 - \alpha}{2} = -1 \Rightarrow \alpha = 6$$